

Digital Image Processing

BCSE 403L AI+TAI

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Q1)

a) $A = [a \ b \ c \ d]$

To find $\Rightarrow$ 4-point DFT of signal A.

$$X_N = W_N u_N, \quad N=4$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

$$W_N = e^{-j\frac{2\pi}{N}}, \quad W_N^N = 1 \Rightarrow W_4^0 = 1$$

$$W_4^1 = e^{-j\frac{2\pi}{4}} = \cos\left(\frac{2\pi}{4}\right) - j\sin\left(\frac{2\pi}{4}\right) = -j$$

$$W_4^2 = -1$$

$$W_4^3 = j$$

$$W_4^4 = 1$$

$$W_4^6 = -1$$

$$W_4^9 = -j$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} a + b + c + d \\ a - bj - c + dj \\ a - b + c - d \\ a + bj - c - dj \end{bmatrix}$$

$$\therefore X(k) = \{ a+b+c+d, a-jb-c+jd, a-b+c-d, a+jb-c-jd \}$$

b) $P = \{ p, t+jv, q, t-jv \}$
 here, $j = \sqrt{-1}$

To find $\Rightarrow$ 4 Point IDFT of P

$$\Rightarrow x_N = \frac{1}{N} W_N^* X_N \quad \therefore N=4$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} W_4^{*0} & W_4^{*0} & W_4^{*0} & W_4^{*0} \\ W_4^{*0} & W_4^{*1} & W_4^{*2} & W_4^{*3} \\ W_4^{*0} & W_4^{*2} & W_4^{*4} & W_4^{*6} \\ W_4^{*0} & W_4^{*3} & W_4^{*6} & W_4^{*9} \end{bmatrix} \begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix}$$

$$W_N^* = e^{j\frac{2\pi}{N}}, \quad W_N^{*0} = 1 \Rightarrow W_N^{*0} = 1$$

$$W_4^{*1} = j$$

$$W_4^{*2} = -1$$

$$W_4^{*3} = -j$$

$$W_4^{*4} = 1$$

$$W_4^{*6} = -j$$

$$W_4^{*9} = j$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} p \\ t+jv \\ q \\ t-jv \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} p+t+jv & +q+t-jv \\ p+jt-v-q & -jt+v \\ p-t-jv & +q-t+jv \\ p-jt+v-q & +jt-v \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} p+2t+q \\ p-q \\ p-2t+q \\ p-q \end{bmatrix}$$

$$\therefore n(x) = \left\{ \frac{p+2t+q}{4}, \frac{p-q}{4}, \frac{p-2t+q}{4}, \frac{p-q}{4} \right\}$$

Q2)

a) $\text{img} = \begin{bmatrix} A & B & C & D \\ E & F & G & H \\ I & J & K & L \\ M & N & O & P \end{bmatrix}$

$$\therefore M=4; N=4$$

2D-DFT Formula:-

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

2D-PFT using Kernel matrix:

For 4×4 ,

$$[K] = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$[F(u,v)] = [K][f(x,y)][K]^T$$

$$F(u,v) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} A & B & C & D \\ E & F & G & H \\ I & J & K & L \\ M & N & O & P \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

⇒

$$\begin{bmatrix} A+E+I+M & B+F+J+N & C+G+K+O & D+H+L+P \\ A-jE-I+jM & B-jF-J+jN & C-jG-K+jO & D-jH-L+jP \\ A-E+I-M & B-F+J-N & C-G+K-O & D-H+L-P \\ A+jE-I-jM & B+jF-J-jN & C+jG-K-jO & D+jH-L-jP \end{bmatrix}$$

$$\times \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

⇒

~~A+B~~ → P

$$\begin{bmatrix} 4A & 4B & 4C & 4D \\ 4M & 4N & 4O & 4P \\ 4I & 4J & 4K & 4L \\ 4E & 4F & 4G & 4H \end{bmatrix}$$

⇒

$$\begin{bmatrix} A & B & C & D \\ E & F & G & H \\ I & J & K & L \\ M & N & O & P \end{bmatrix}$$

⇒

$$\begin{bmatrix} A & B & C & D \\ M & N & O & P \\ I & J & K & L \\ E & F & G & H \end{bmatrix}$$

$$b) \text{IPFT}[F(u,v)] \Rightarrow f(x,y)$$

$$f(x,y) = \frac{1}{MN} [K]^T [F] [K]$$

$$F(u,v) = \begin{bmatrix} 4A & 4B & 4C & 4D \\ 4M & 4N & 4O & 4P \\ 4I & 4J & 4K & 4L \\ 4E & 4F & 4G & 4H \end{bmatrix}$$

$$\frac{1}{16} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} A & B & C & D \\ M & N & O & P \\ I & J & K & L \\ E & F & G & H \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$\Rightarrow \frac{1}{4} \begin{bmatrix} A+M+I+E & B+N+J+F & C+O+K+G & D+P+L+H \\ A-jM-I+jE & B-jN-J+jF & C-jO-K+jG & D-jP-L+jH \\ A-M+I-E & B-N+J-F & C-O+K-G & D-P+L-H \\ A+jM-I-jE & B+jN-J-jF & C+jO-K-jG & D+jP-L-jH \end{bmatrix}$$

$$\times \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$\Rightarrow \frac{1}{4} \begin{bmatrix} 4A & 4B & 4C & 4D \\ 4E & 4F & 4G & 4H \\ 4I & 4J & 4K & 4L \\ 4M & 4N & 4O & 4P \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} A & B & C & D \\ E & F & G & H \\ I & J & K & L \\ M & N & O & P \end{bmatrix}$$

Q3)

1) if $f(x, y)$ is real, then $F(u, v) = F^*(-u, -v)$

in DFT

$$f(x, y) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j 2\pi (ux/M + vy/N)}$$

$$f^*(u, v) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{j 2\pi (ux/M + vy/N)}$$

Substituting $u \rightarrow -u$ and $v \rightarrow -v$

$$\begin{aligned} F(-u, -v) &= \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{j 2\pi (ux/M + vy/N)} \\ &= F^*(u, v) \end{aligned}$$

Thus if $f(x, y)$ is real, $F(u, v)$ has Hermitian Symmetry

1) if $f(x,y)$ is imaginary then $F(-u, -v) = -F^*(u, v)$

Since $f(x,y)$ is imaginary, let $f(x,y) = jg(x,y)$ where $g(x,y)$ is a real function. Taking the DFT:

$$F(u, v) = jG(u, v)$$

Where $G(u, v)$ is the Fourier Transform of $g(x,y)$, a real function. ~~Taking~~ From Property 1, we know that

$$G(-u, -v) = G^*(u, v), \text{ so:}$$

$$F(-u, -v) = jG(-u, -v) = jG^*(u, v) = -F^*(u, v)$$

3) if $f(x,y)$ is real, then $R(u, v)$ is even and $I(u, v)$ is odd

Here, $R(u, v)$ and $I(u, v)$ are the real and imaginary parts of $F(u, v)$

Proof:

From Property 1

$$F(u, v) = F^*(-u, -v) = R(-u, -v) - jI(-u, -v)$$

Equating real and imaginary parts:

$$R(u, v) = R(-u, -v) \text{ (even)}$$

$$I(u, v) = -I(-u, -v) \text{ (odd)}$$

4) if $f(x,y)$ is imaginary, then $R(u, v)$ is odd and $I(u, v)$ is even

From Property 2 we know:

$$F(-u, -v) = -F^*(u, v) = -R(u, v) - j(-I(u, v))$$

equating real and imaginary parts:

$$R(u,v) = -R(-u, -v) \text{ (odd)}$$

$$I(u,v) = I(-u, -v) \text{ (even)}$$

- 5) If $f(x,y)$ is real and shifted ($f(x,y) \rightarrow f(-x, -y)$), then $F(u,v)$ becomes complex-conjugate symmetric $F^*(u,v)$

From Property 1, we know that shifting results in phase changes, making the transform complex.

- 6) if $f(x,y)$ is complex, then $F(u,v)$ is also complex

Since, $f(x,y)$ has both real and imaginary parts, the transform $F(u,v)$ will have no special symmetry.

- 7) if $f(x,y)$ is complex conjugate $f^*(x,y)$, then $F(u,v)$ is complex

Since conjugation in the spatial domain results in conjugation in the frequency domain, $F(u,v)$ remains complex.

- 8) If $f(x,y)$ is real and even, then $F(u,v)$ is real and even.

if $f(x,y) = f(-x, -y)$, then using DFT symmetry properties

$$F(u,v) = \sum f(x,y) e^{-j2\pi(ux/M + vy/N)}$$

Since exponentials remain symmetric, $F(u,v)$ remains real and even.

9) if $f(x,y)$ is real and odd, then $F(u,v)$ is imaginary and odd

if $f(x,y) = -f(-x,-y)$, then plugging into the DFT equation leads to $F(u,v)$ being purely imaginary

10) If $f(x,y)$ is imaginary and even, then $F(u,v)$ is imaginary and even

if $f(x,y) = -f^*(x,y)$ and symmetric, the transform remains imaginary

11) if $f(x,y)$ is imaginary and odd, then $F(u,v)$ is real and odd

→ Using conjugation and symmetry properties, it follows that $F(u,v)$ is real and odd

12) If $f(x,y)$ is complex and even, then $F(u,v)$ is complex and even

→ A symmetric complex function transforms into a symmetric complex frequency domain representation

13) If $f(x,y)$ is complex and odd, then $F(u,v)$ is complex and odd

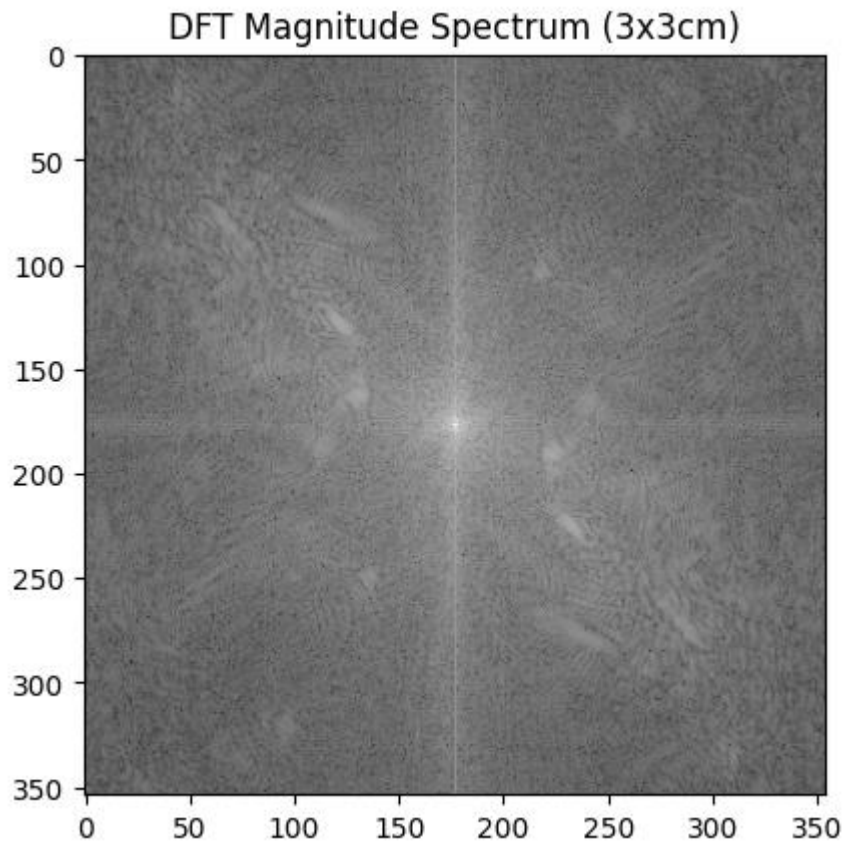
→ Since the input is fully asymmetric, the output remains asymmetric in frequency space

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread("manis.jpg", cv2.IMREAD_GRAYSCALE)
cv2.imshow("Resized Image", resized_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
dpi = 300
cm_to_pixels = lambda cm: int(cm * dpi / 2.54)
resized_img = cv2.resize(img, (cm_to_pixels(3), cm_to_pixels(3)),
interpolation=cv2.INTER_AREA)
dft = cv2.dft(np.float32(resized_img), flags=cv2.DFT_COMPLEX_OUTPUT)
dft_shift = np.fft.fftshift(dft)
magnitude_spectrum = 20 * np.log(cv2.magnitude(dft_shift[:, :, 0],
dft_shift[:, :, 1]))
plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("DFT Magnitude Spectrum (3x3cm)")
plt.show()
```

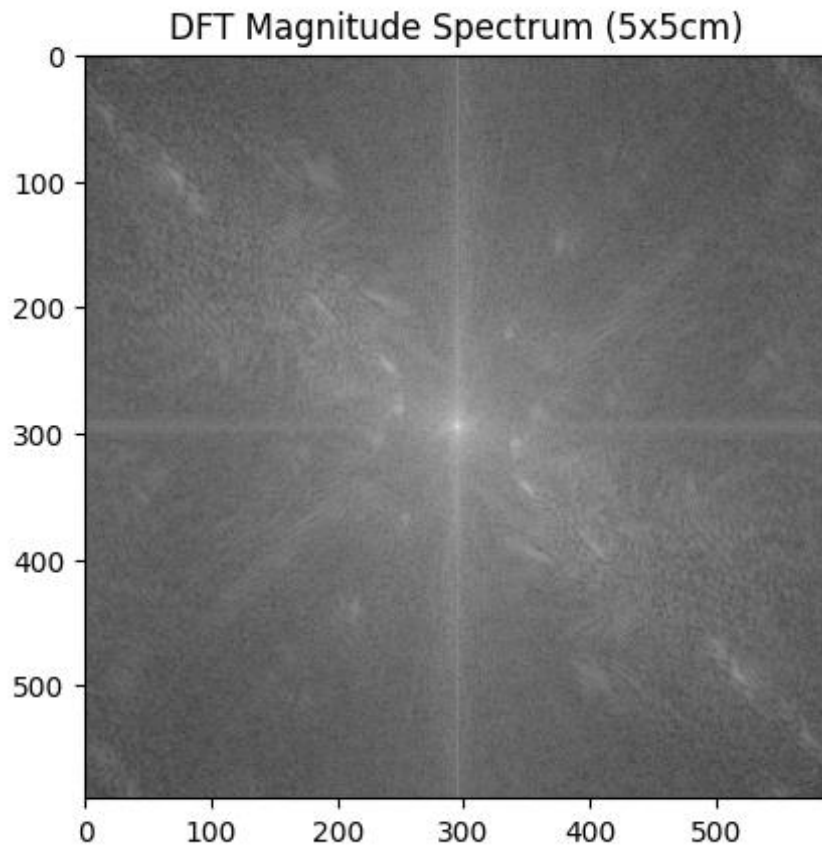




Code:

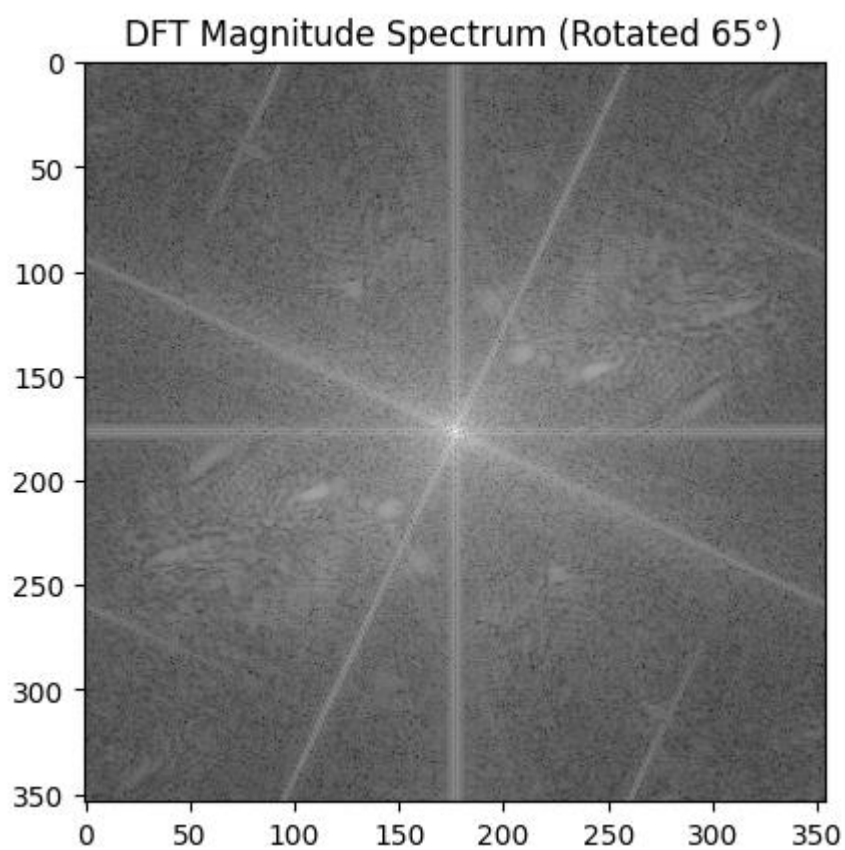
```
resized_img_5cm = cv2.resize(img, (cm_to_pixels(5), cm_to_pixels(5)),
interpolation=cv2.INTER_AREA)

dft_5cm = cv2.dft(np.float32(resized_img_5cm), flags=cv2.DFT_COMPLEX_OUTPUT)
dft_shift_5cm = np.fft.fftshift(dft_5cm)
magnitude_spectrum_5cm = 20 * np.log(cv2.magnitude(dft_shift_5cm[:, :, 0],
dft_shift_5cm[:, :, 1]))
plt.imshow(magnitude_spectrum_5cm, cmap='gray')
plt.title("DFT Magnitude Spectrum (5x5cm)")
plt.show()
```



Code:

```
center = (resized_img.shape[1] // 2, resized_img.shape[0] // 2)
rotation_matrix = cv2.getRotationMatrix2D(center, 65, 1.0)
rotated_img = cv2.warpAffine(resized_img, rotation_matrix,
                              (resized_img.shape[1], resized_img.shape[0]))
dft_rotated = cv2.dft(np.float32(rotated_img),
                      flags=cv2.DFT_COMPLEX_OUTPUT)
dft_shift_rotated = np.fft.fftshift(dft_rotated)
magnitude_spectrum_rotated = 20 *
np.log(cv2.magnitude(dft_shift_rotated[:, :, 0], dft_shift_rotated[:, :,
1])))
plt.imshow(magnitude_spectrum_rotated, cmap='gray')
plt.title("DFT Magnitude Spectrum (Rotated 65°)")
plt.show()
```

DA2

4. Prove properties :-

a) Average value $\bar{f} = \frac{1}{MN} \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(n,y) = \frac{1}{MN} F(0,0)$

b) Periodicity (k_1 and k_2 are integers)

$$F(u,v) = F(u+k_1M, v) = F(u, v+k_2N) = F(u+k_1M, v+k_2N)$$

$$f(n,y) = f(n+k_1M, y) = f(n, y+k_2N) = f(n+k_1M, y+k_2N)$$

c) Separability The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result.

d) Obtaining the IDFT using a DFT algorithm

$$MN f^*(n,y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u,v) e^{-j2\pi(ux/M + vy/N)}$$

This equation indicates the inputting $F^*(u,v)$ into an algorithm that computes the forward transform (right side of the above equation) yields $MN f^*(n,y)$. Taking the complex conjugate and dividing by MN gives the desired inverse.

Ans a) The average value of a digital image $f(n,y)$ of size $M \times N$ is given by:

$$\bar{f} = \frac{1}{MN} \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(n,y)$$

An image is represented as a discrete function $f(n,y)$ where n and y are spatial coordinates.

The mean intensity $\bar{f}$ represents the average pixel intensity across the entire image.

The total sum of all pixels values in an image of dimensions $M \times N$ is:

$$S = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y)$$

The average intensity $\bar{f}$ is given by dividing this sum by the total number of pixels MN :

$$\bar{f} = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y)$$

This formula simply states that the average intensity of an image is obtained by summing all pixel values and normalizing by the number of pixels.

It ensures that $\bar{f}$ represents the overall brightness of the image.

For a continuous image function $f(x,y)$ over a region $[0, M] \times [0, N]$, the continuous mean intensity would be given by:

$$\bar{f} = \frac{1}{MN} \int_0^M \int_0^N f(x,y) dy dx$$

b) The periodicity property states that a discrete image function $f(x,y)$ exhibits periodic behavior in both the spatial domain and the frequency domain, that is, if k_1 and k_2 are integers:

$$f(x+k_1M, y+k_2N) = f(x,y)$$

Similarly, for the Fourier transform $F(u,v)$ of $f(x,y)$:

$$F(u+k_1M, v+k_2N) = F(u,v)$$

This means that the Fourier Transform of an image is periodic M along u and N along v .

The 2D DFT of an image $f(x,y)$ of size $M \times N$ is given by:

$$F(u, v) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(n, y) e^{-j2\pi \left(\frac{un}{M} + \frac{vy}{N} \right)}$$

where j is the imaginary unit ($j^2 = -1$),

u and v are the frequency coordinates.

The inverse 2D DFT is given by.

$$f(n, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi \left(\frac{un}{M} + \frac{vy}{N} \right)}$$

These equations transform the spatial image $f(n, y)$ into its frequency representation $F(u, v)$ and vice versa.

$$\begin{aligned} f(n+k_1M, y+k_2N) &= \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi \left(\frac{(n+k_1M)u}{M} + \frac{(y+k_2N)v}{N} \right)} \\ &= \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi \left(\frac{un}{M} + \frac{vy}{N} \right)} e^{j2\pi (vk_1 + vk_2)} \end{aligned}$$

Since u and v are integers, the term $e^{j2\pi n}$ simplifies to 1 (by Euler's formula, because $e = 1$ for any integer n).

$$f(n+k_1M, y+k_2N) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi \left(\frac{un}{M} + \frac{vy}{N} \right)}$$

Hence, $f(n+k_1M, y+k_2N) = f(n, y)$.

Similarly, consider shifting u by k_1M and v by k_2N .

$$F(u+k_1M, v+k_2N)$$

Using the DFT definition

$$\begin{aligned} F(u+k_1M, v+k_2N) &= \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(n, y) e^{-j2\pi \left(\frac{(u+k_1M)n}{M} + \frac{(v+k_2N)y}{N} \right)} \\ &= \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(n, y) e^{-j2\pi \left(\frac{un}{M} + \frac{vy}{N} \right)} e^{-j2\pi (k_1n + k_2y)} \end{aligned}$$

Since n and y are integers, $e^{-j2\pi (k_1n + k_2y)} = 1$, so we get

$$F(u+k_1M, v+k_2N) = F(u, v)$$

Thus, the Fourier Transform is also periodic in both frequency dimensions.

c) The 2D DFT of an image $f(x,y)$ of size $M \times N$ is given by: $F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$

This can be rewritten as a two-step process:

1. Compute the 1D DFT along rows:

Define an intermediate function $G(u,y)$ by applying the 1DFT along rows:

$$G(u,y) = \sum_{x=0}^{M-1} f(x,y) e^{-j2\pi \frac{ux}{M}}$$

2. Compute the 1D DFT along columns:

Now, apply the 1D DFT along columns to $G(u,y)$:

$$F(u,v) = \sum_{y=0}^{N-1} G(u,y) e^{-j2\pi \frac{vy}{N}}$$

Thus, the 2DFT is separable into two consecutive 1D DFTs, making computation efficient, especially in FFT-based implementation.

d) The 2D DFT is given by:

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

The inverse 2D DFT (IDFT) is:

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

Taking the complex conjugate of $F(u,v)$:

$$F^*(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f^*(x,y) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

Now, apply the forward DFT on $F^*(u,v)$:

$$\sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u,v) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} = MN f^*(x,y)$$

Taking the complex conjugate of the result:

$$(MN f^*(u, v))^* = MN f(u, v)$$

Dividing by MN , we obtain the inverse transform:

$$f(u, v) = \frac{1}{MN} \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} F(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

5. Prove properties:-

a) Linearity $af_1(u, v) + bf_2(u, v) \Leftrightarrow aF_1(u, v) + bF_2(u, v)$

b) Translation (general) $f(u, v) e^{j2\pi(u_0x/M + v_0y/N)} \Leftrightarrow F(u - u_0, v - v_0)$
 $f(u - u_0, v - v_0) \Leftrightarrow F(u, v) e^{j2\pi(\frac{u_0x}{M} + \frac{v_0y}{N})}$

c) Translation to centre of the frequency rectangle $f(u, v) (-1)^{u+1} \Leftrightarrow F(u - M/2, v - N/2)$
 $f(u - M/2, v - N/2) \Leftrightarrow F(u, v) (-1)^{u+1}$

d) Rotation $f(u, v) \Leftrightarrow F(u, v)$
 $r = \sqrt{u^2 + v^2} \quad \theta = \tan^{-1}(v/u) \quad \omega = \sqrt{u^2 + v^2} \quad \phi = \tan^{-1}(v/u)$

e) Discrete unit impulse $\delta(u, v) \Leftrightarrow 1$
 $1 \Leftrightarrow MN \delta(u, v)$

f) Rectangle $\text{rec}[a, b] \Leftrightarrow ab \frac{\sin(\pi ua)}{(\pi ua)} \frac{\sin(\pi vb)}{(\pi vb)} e^{-j\pi(ua+vb)}$

g) Sine $\sin(2\pi u_0x/M + 2\pi v_0y/N) \Leftrightarrow \frac{1}{2} [F(u + u_0, v + v_0) - F(u - u_0, v - v_0)]$

h) Cosine $\cos(2\pi u_0x/M + 2\pi v_0y/N) \Leftrightarrow \frac{1}{2} [F(u + u_0, v + v_0) + F(u - u_0, v - v_0)]$

Ans a) Consider two functions $f_1(u, v)$ and $f_2(u, v)$ with their respective DFTs $F_1(u, v)$ and $F_2(u, v)$

$$F_1(u, v) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f_1(u, v) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

$$F_2(u, v) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f_2(u, v) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

Now, consider a linear combination $g(u, v) = af_1(u, v) + bf_2(u, v)$

Taking the DFT of $g(x,y)$:

$$G(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [a f_1(x,y) + b f_2(x,y)] e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

Using linearity of summation:

$$G(u,v) = a \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f_1(x,y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} +$$

$$b \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f_2(x,y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

$$G(u,v) = a F_1(u,v) + b F_2(u,v).$$

b) Consider shifting the function by (x_0, y_0) :

$$g(x,y) = f(x-x_0, y-y_0)$$

Taking the DFT of $g(x,y)$:

$$G(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x-x_0, y-y_0) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

let $x' = x - x_0$ and $y' = y - y_0$, so that $x = x' + x_0$

and $y = y' + y_0$. Substituting these:

$$G(u,v) = \sum_{x'=0}^{M-1} \sum_{y'=0}^{N-1} f(x', y') e^{-j2\pi(\frac{u(x'+x_0)}{M} + \frac{v(y'+y_0)}{N})}$$

$$G(u,v) = \sum_{x'=0}^{M-1} \sum_{y'=0}^{N-1} f(x', y') e^{-j2\pi(\frac{ux'}{M} + \frac{vy'}{N})} e^{-j2\pi(\frac{ux_0}{M} + \frac{vy_0}{N})}$$

$$G(u,v) = e^{-j2\pi(\frac{ux_0}{M} + \frac{vy_0}{N})} \sum_{x'=0}^{M-1} \sum_{y'=0}^{N-1} f(x', y') e^{-j2\pi(\frac{ux'}{M} + \frac{vy'}{N})}$$

Since the summation is just the original DFT $F(u,v)$, we obtain:

$$G(u,v) = e^{-j2\pi(\frac{ux_0}{M} + \frac{vy_0}{N})} F(u,v)$$

c) The goal is to shift the zero frequency (DC component) from $(0,0)$ to (Mh, Nh) . This is done by multiplying the spatial function $f(n,y)$ by $(-1)^{ny}$.

$$g(n,y) = (-1)^{ny} f(n,y)$$

Taking the DFT to $g(n,y)$:

$$G(u,v) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} (-1)^{ny} f(n,y) e^{-j2\pi \left(\frac{un}{M} + \frac{vy}{N} \right)}$$

$$(-1)^{ny} = e^{j\pi(ny)}$$

$$G(u,v) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(n,y) e^{j\pi(ny)} e^{-j2\pi \left(\frac{un}{M} + \frac{vy}{N} \right)}$$

$$G(u,v) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(n,y) e^{-j2\pi \left(\frac{un}{M} + \frac{vy}{N} - \frac{n}{2} - \frac{y}{2} \right)}$$

$$G(u,v) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(n,y) e^{-j2\pi \left(\frac{(u-Mh)n}{M} + \frac{(v-Nh)y}{N} \right)}$$

which is the DFT of $f(n,y)$ evaluated at $(u-Mh, v-Nh)$.

d) If we rotate the image $f(n,y)$ by an angle θ , we obtain a new function $g(n',y')$, where:

$$n' = x \cos \theta + y \sin \theta \quad y' = -x \sin \theta + y \cos \theta$$

Thus, the rotated function is: $g(n',y') = f(n,y)$

The DFT of $g(n',y')$ is:

$$G(u',v') = \sum_{n',y'} g(n',y') e^{-j2\pi \left(\frac{u'n'}{M} + \frac{v'y'}{N} \right)}$$

$$G(u',v') = \sum_{n,y} f(n,y) e^{-j2\pi \left(\frac{u'(x \cos \theta + y \sin \theta)}{M} + \frac{v'(-x \sin \theta + y \cos \theta)}{N} \right)}$$

$$G(u',v') = \sum_{n,y} f(n,y) e^{-j2\pi \left(\frac{(u' \cos \theta - v' \sin \theta)x}{M} + \frac{(u' \sin \theta + v' \cos \theta)y}{N} \right)}$$

Recognizing this as the DFT of $f(n,y)$ evaluated at the rotated frequency coordinates: $G(u',v') = F(u,v)$

where $u = u' \cos \theta - v' \sin \theta$, $v = u' \sin \theta + v' \cos \theta$

e) The 2D discrete unit impulse function (also known as the Kronecker delta) is defined as:

$$\delta(n, y) = \begin{cases} 1, & \text{if } n=0, y=0 \\ 0, & \text{otherwise.} \end{cases}$$

By definition, the 2D DFT is:

$$F(u, v) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} \delta(n, y) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

Since $\delta(n, y)$ is 1 only at $(0, 0)$ and 0 elsewhere, the summation simplifies to:

$$F(u, v) = e^{j2\pi \left(\frac{0+u}{M} + \frac{0+v}{N} \right)}$$

$$F(u, v) = 1$$

Thus the DFT of a unit impulse function in the spatial domain results in a constant value 1 in the frequency domain:

$$\delta(n, y) \xrightarrow{F} 1$$

The inverse 2DDFT is given by:

$$f(n, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} 1 \cdot e^{j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

Since the summation over all exponentials of the form $e^{j2\pi k}$ for integer $k \neq 0$ results in zero, and only the term at $(n, y) = (0, 0)$ remains non-zero, we get:

$$f(n, y) = \frac{1}{MN} \cdot MN \delta(n, y) = \delta(n, y)$$

Thus, the inverse DFT of a constant function results in the unit impulse function

$$1 \xrightarrow{F^{-1}} \frac{1}{MN} \delta(n, y)$$

f) A rectangular function of width $2a$ and height $2b$ is given by:

$$\text{rect}(x/a, y/b) = \begin{cases} 1, & \text{if } |x| \leq a, |y| \leq b \\ 0, & \text{otherwise} \end{cases}$$

The 2D Discrete Fourier Transform (DFT) is defined as:

$$F(u, v) = \sum_{x=-a}^a \sum_{y=-b}^b e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

Since the rectangular function is non-zero only within $|x| \leq a$ and $|y| \leq b$, the summation simplifies to:

$$F(u, v) = \sum_{x=-a}^a e^{-j2\pi \frac{ux}{M}} \sum_{y=-b}^b e^{-j2\pi \frac{vy}{N}}$$

Each sum is a geometric series, which has a known closed-form solution:

$$\sum_{k=-M}^M e^{-j2\pi \frac{mk}{M}} = e^{-j2\pi \frac{2a}{M} u} \frac{\sin((2a+1)\frac{\pi u}{M})}{\sin(\frac{\pi u}{M})}$$

Applying this to both x and y directions, we get:

$$F(u, v) = e^{-j2\pi \frac{au}{M}} \frac{\sin((2a+1)\frac{\pi u}{M})}{\sin(\frac{\pi u}{M})} \cdot e^{-j2\pi \frac{bv}{N}} \frac{\sin((2b+1)\frac{\pi v}{N})}{\sin(\frac{\pi v}{N})}$$

Thus, the 2D DFT of a rectangular function results in a sinc function in both frequency dimensions:

$$\text{rect}(x/a, y/b) \xrightarrow{\text{DFT}} e^{-j2\pi \frac{au}{M}} \frac{\sin((2a+1)\frac{\pi u}{M})}{\sin(\frac{\pi u}{M})} \cdot e^{-j2\pi \frac{bv}{N}} \frac{\sin((2b+1)\frac{\pi v}{N})}{\sin(\frac{\pi v}{N})}$$

g) A sine wave in two dimensions is generally expressed as:

$$f(x, y) = \sin\left(2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N}\right)\right)$$

where u_0 and v_0 are the frequency components in the x and y directions, and M, N are the image dimensions.

The 2D DFT is given by:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \sin\left(2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N}\right)\right) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Using Euler's formula: $\sin(A) = \frac{e^{jA} - e^{-jA}}{2j}$

We can rewrite the sine function as:

$$f(u, v) = \frac{1}{2j} \left(e^{j2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)} - e^{-j2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)} \right)$$

Substituting into the DFT equation:

$$P(u, v) = \frac{1}{2j} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \left(e^{j2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)} - e^{-j2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)} \right) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

Expanding the exponentials:

$$P(u, v) = \frac{1}{2j} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \left(e^{j2\pi \left(\frac{(u_0-u)x}{M} + \frac{(v_0-v)y}{N} \right)} - e^{-j2\pi \left(\frac{(u_0+u)x}{M} + \frac{(v_0+v)y}{N} \right)} \right)$$

Using the DFT property of discrete delta functions, the summation results in:

$$P(u, v) = \frac{MN}{2j} (\delta(u-u_0, v-v_0) - \delta(u+u_0, v+v_0))$$

where $\delta(u, v)$ is the discrete unit impulse function.

$$\sin\left(2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)\right) \xleftrightarrow{F} \frac{MN}{2j} [\delta(u-u_0, v-v_0) - \delta(u+u_0, v+v_0)]$$

b) A cosine wave in two dimension is given by:

$$f(u, v) = \cos\left(2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)\right)$$

where u_0 and v_0 are the frequency components in the x and y directions, and M, N are the image dimensions.

The 2D DFT is given by:

$$P(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \cos\left(2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)\right) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

Using Euler's formula: $\cos(A) = \frac{e^{jA} + e^{-jA}}{2}$

we can rewrite the cosine function as:

$$A(u, v) = \frac{1}{2} \left(e^{j2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)} + e^{-j2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)} \right)$$

Substituting into the DFT equation:

$$F(u, v) = \frac{1}{2} \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} \left(e^{j2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)} + e^{-j2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right)} \right)$$

Expanding the exponentials:

$$F(u, v) = \frac{1}{2} \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} \left(e^{j2\pi \left(\frac{(u-u_0)x}{M} + \frac{(v-v_0)y}{N} \right)} + e^{-j2\pi \left(\frac{(u-u_0)x}{M} + \frac{(v-v_0)y}{N} \right)} \right)$$

Using the DFT property of discrete delta functions, the summation results in:

$$F(u, v) = \frac{MN}{2} \left(\delta(u-u_0, v-v_0) + \delta(u+u_0, v+v_0) \right)$$

where $\delta(u, v)$ is the discrete unit impulse function.

$$\cos \left(2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right) \right) \xleftrightarrow{F} \frac{MN}{2} \left[\delta(u-u_0, v-v_0) + \delta(u+u_0, v+v_0) \right]$$

6. Study and explain in three pages the technique of 'Fast Fourier Transform' for computing DFT. Explain the techniques with the help of 'Butterfly diagram' for 8-point 1-D DFT. Explain how much speedup happens in this technique.

Ans - The Fast Fourier Transform (FFT) is an efficient algorithm used to compute the Discrete Fourier Transform (DFT) and its inverse. The DFT of a sequence of N complex numbers is given by:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}, \quad k = 0, 1, 2, \dots, N-1$$

The direct computation of the DFT requires

$O(N^2)$ operations. However, the FFT reduces this complexity to $O(N \log N)$ using a divide-and-conquer strategy. This makes FFT significantly faster, especially for large values of N .

The most common FFT algorithm is the Radix-2 Decimation-in-Time (DIT) FFT, which applies when N is a power of 2. It works as follows:

The sequence $x(n)$ is split into even and odd indexed terms:

$$X(k) = \sum_{n=0}^{N/2-1} x(2n) e^{-j2\pi k(2n)/N} + \sum_{n=0}^{N/2-1} x(2n+1) e^{-j2\pi k(2n+1)/N}$$

By separating the even and odd terms, we can rewrite this as:

$$X(k) = X_{\text{even}}(k) + e^{j2\pi k/N} X_{\text{odd}}(k)$$

This step recursively divides the problem into smaller FFTs of size $N/2$, reducing the overall computation.

Using symmetry properties of the exponential term (Twiddle Factor): $W_N^k = e^{-j2\pi k/N}$

we compute $X(k) = X_{\text{even}}(k) + W_N^k X_{\text{odd}}(k)$

$$X(k + N/2) = X_{\text{even}}(k) - W_N^k X_{\text{odd}}(k)$$

This is known as the Butterfly operation, and it

reduces the computation significantly.

An 8-point FFT follows these stages:

1. Bit-reversal permutation: The input sequence is reordered based on the bit-reversed index.
 2. Recursive computation: Pairs of elements are combined using the Butterfly operation.
- a. Given an input sequence $x(0), x(1), \dots, x(7)$, we reorder it using bit-reversal.

Original index	Binary	Bit-Reversed Index
0	000	000 (0)
1	001	100 (4)
2	010	010 (2)
3	011	110 (6)
4	100	001 (1)
5	101	101 (5)
6	110	011 (3)
7	111	111 (7)

b. Each stage consists of Butterfly operations.

c. Pairwise computations are performed between values spaced by 4.

d. Intermediate results are combined with appropriate twiddle factors.

e. Final computation gives $X(k)$, the DFT output.

The speed up ratio can be approximated as:

$$\frac{O(N^2)}{O(N \log N)} = \frac{N}{\log N}$$

N	N^2	$N \log N$	Speed up Ratio.
8	64	24	2.67x
16	256	64	4x
1024	1048576	10240	102.4x